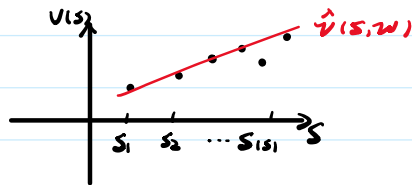


Continuous (s, a) ?eg. 1D states: $s_1, s_2, \dots, s_{|S|}$.

$$\Rightarrow V_{\pi}(s_1), V_{\pi}(s_2), \dots, V_{\pi}(s_{|S|})$$



$$\hat{v}(s, w) = as + b = (s \ 1) \begin{pmatrix} a \\ b \end{pmatrix} = \phi^T(s)w$$

feature vector of $s: (s, 1)$ also: $(s^2, s, 1) \dots \Rightarrow \text{nonlinearity}$

State Value Approximation

① Objective function:

$$J(w) = E[(V_{\pi}(s) - \hat{v}(s, w))^2]$$

uniform distribution:

$$J(w) = \frac{1}{|S|} \sum_{s \in S} (V_{\pi}(s) - \hat{v}(s, w))^2$$

stationary distribution:

$$J(w) = \sum_{s \in S} d_{\pi}(s) (V_{\pi}(s) - \hat{v}(s, w))^2$$

?

$$\text{method 1: } d_{\pi}(s) = \frac{n_{\pi}(s)}{\sum_{s' \in S} n_{\pi}(s')} \rightarrow \text{episode length}$$

$$\text{method 2: } d_{\pi}^T = d_{\pi}^T P_{\pi}$$

$$\begin{aligned} \text{② Optimizer: } D_w J(w) &= E[D_w (V_{\pi}(s) - \hat{v}(s, w))^2] \\ &= 2E[(V_{\pi}(s) - \hat{v}(s, w))(-D_w \hat{v}(s, w))] \end{aligned}$$

$$\Rightarrow \text{SAD: } w_{t+1} = w_t + \alpha_t (V_{\pi}(s_t) - \hat{v}(s_t, w_t)) D_w \hat{v}(s_t, w_t)$$

$$+ \text{MC: } w_{t+1} = w_t + \alpha_t (g_t - \hat{v}(s_t, w_t)) D_w \hat{v}(s_t, w_t)$$

$$+ \text{TD: } w_{t+1} = w_t + \alpha_t (r_{t+1} + \gamma \hat{v}(s_{t+1}, w_t) - \hat{v}(s_t, w_t)) D_w \hat{v}(s_t, w_t)$$

actually minimize Bellman Error. / Projected Bellman Error

Sarsa with function approximation

Algo (Sarsa + fa)

for episode. do:

$$\pi_t(s_t) \rightarrow a_t \quad \& \quad r_{t+1}, s_{t+1}, \pi_t(s_{t+1}) \rightarrow a_{t+1}$$

Value update (w-update):

$$w_{t+1} = w_t + \alpha_t [r_{t+1} + \gamma \hat{Q}(s_{t+1}, a_{t+1}, w_t) - \hat{Q}(s_t, a_t, w_t)] \nabla_w \hat{Q}(s_t, a_t, w_t)$$

policy update:

$$\pi_{t+1}(a|s_t) = \begin{cases} 1 - \frac{\epsilon}{|A(s_t)|} (|A(s_t)| - 1) & , a = \arg\max_a \hat{Q}(s_t, a, w_{t+1}) \\ \frac{\epsilon}{|A(s_t)|} & , \text{else} \end{cases}$$

Q-learning with function approximation

...

* DQN (deep Q-network)

$$J = E \left[(r + \gamma \max_{a \in A(s')} \hat{Q}(s', a, w_T) - \hat{Q}(s, A, w))^2 \right]$$

{ main network
target network. (w_T fixed periodically).

$$\Rightarrow \nabla_w J = E \left[(r + \gamma \max_{a \in A(s')} \hat{Q}(s', a, w_T) - \hat{Q}(s, A, w)) \nabla_w \hat{Q}(s, A, w) \right]$$

Algo (DQN)

Init: main network + target network (same structure, same init).

$$\pi_0 \rightarrow (s_0, a_0, r_1, s_1, a_1, r_2, s_2, \dots)$$

Replay Buffer (Experience Replay): $\mathcal{B} = \{(s, a, r, s')\}$

for each iteration, do:

- ① draw a mini-batch (uniformly) from $\mathcal{B}$.
- ② $(s, a, r, s') \rightarrow \underline{y_T = r + \gamma \max_{a' \in A(s')} \hat{Q}(s', a', w_T)}$ target value
- ③ main network: minimize $(y_T - \hat{Q}(s, a, w))^2$.

set $w_T = w$ every C iterations.

